Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Structure-property relationship

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Structure-property relationship
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Structure-property relationship

1 Structure-property relationship

Study focus: Structure-property relationship. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000078 · EV0000079

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials, that is, polymers.

Key details

- We will talk about the structure property relationship.
- What do we understand by the term the structure property relationship?
- The properties of polymers are influenced by their structure.
- Polymers when it comes to properties are extremely versatile.
- Some can be hard, whereas others can be very soft and flexible.
- Some could be insoluble and resistant to heat, while others could be soluble and fusible.
- But here, normally the polymer will be in the form of this amorphous matrix.
- But in some polymers, we find that within this matrix also there could be some alignment like this where the polymer chains come very close to each other.
- Otherwise if those are missing then that is going to be amorphous in nature.
- Are the chains coming close to each other, is it possible for the chains to come close to each other and that would discur decide the crystallinity in a polymer.
- What could be the reason for a polymer to be crystalline?
- It depends on the structure to a great extent.
- CH2, CH2, CH2.
- combine each of these is CH2, CH2, CH2.
- Like this you have all these chains and since there is no obstruction for the chains to come close to each other we can say that this will be more crystalline as compared to a polyvenyl acetate.
- Polyvinyl acetate would be like this CH.
- O C O C H three.

- C H O C O.
- So you can imagine there will be another chain like this, and therefore these groups are referred to as bulky pendant groups.
- When you have such bulky pendant groups, obviously there will be an obstruction for the chains to come close to each other.
- Therefore such a polymer would definitely be less crystalline or more amorphous as compared to when there are no bulky pendant groups.
- Second reason is the presence of polar groups.
- Whenever there are polar groups, then it is found the polymer becomes more crystalline.
- Because whenever you have these polar groups in the.
- Polymer chain, we know that this could give rise to dipole interactions or hydrogen bonding.
- Polyvinyl fluoride, we know that the chain is like this, extending.
- And similarly you will have another chain here.
- So we find that because of the polar nature.
- We know that there is polarity here, H gets a small positive, and this bond is going to slightly shift here.
- So therefore, there is attraction between the between the chains.
- There is a dipole interaction which is happening between the chains, and therefore the chains can come closer whereas in polyethylene it is just as we saw just now CH2 and there is no such attraction except for the Van der Waals forces of attraction.
- Therefore we find that in the case of polyethylene it is.

Example or application from the lecture

- The properties we consider will consider will be crystallinity, tensile strength, elasticity, chemical resistance and plastic deformation.
- Let us take up the first property, crystallinity.
- For example, polyethylene.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 05:00–09:59

much the crystallinity will be less as compared to polynomial fluoride.

Key details

- The crystallinity also depends on configuration.
- Ste is that is stereoregularity, crystallinity.
- When we say configuration, we are talking of stereoregularity, that is how are the substituents arranged in space.
- Here in syndioctic they are occupying alternate positions, whereas in A-tactic, they are just occupying any random position.
- The polymer is found to be most crystalline, whereas in A- tactic it is found to be least crystalline.
- What is the reason?

- Crystallin T is because you are the chains can come close to each other.
- We can see in isotactic they can come much closer to each other as compared to syntyotactic and a tactic.
- Coming to the next property, tensile strength.
- What would be tensile strength?
- The amount of stress a polymer can take before undergoing permanent deformation.
- By increasing the molecular weight we can understand that the tensile strength is going to increase.
- Like you can see in this graph, the tensile strength, when you increase the degree of polymerization or molecular weight, the tensile strength is increasing to a limit, and then after that it becomes almost constant.
- Increasing further would not make too much difference in the tensile stem.
- But one more property which we have to look at is the melt viscosity.
- As we increase the degree of polymerization of the molecular weight, the melt viscosity also goes on increasing.
- Remember that polymer.
- Something which we just test in the lab, they are for commercial purpose.
- Therefore, we need to make a compromise and therefore somewhere in this region where the tensile strength is also high.
- And the valve viscosity is not very high.
- This is the range which is used for fabrication of the polymers, I mean manufacture of the polymers, preparation of the polymers.
- PDC just now we know that there is a dipole attraction between the molecules, and when there is dipole attraction, obviously the chains will not slip each other slip against each other.
- Whereas like in the case of polyethylene, what did we notice in polyethylene?
- I just draw like this.
- This is the carbon chain which we have CH2, CH2, CH2.
- Because there's just Van der Waals forces.
- Tensile strength when you're applying a force on it, the chains will slip against each other.
- Whereas if you have some attraction, like in the case of nylon, between the chains we find that there is hydrogen bonding, there is a attraction that is.
- Hydrogen bonding between the chains.
- And of course, if we have cross linking, then.
- Definitely the inner polymer will also increase with the molecular weight and therefore there will be increase in the tensile strength.
- So therefore when we have cross linking like this, we find.
- Cross linking means there is covalent bond formation between the chains network three dimension network and therefore definitely it is going to this is going to result in increase in tensile strength.
- Cross linking also increases the tensile strength.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is the next property we need to consider.

Key details

- The property of a polymer by which it deforms on applying force and regains its original shape when we when the force is released, and that is known as elasticity.
- There is a.
- it goes to the structure again.
- They have long coiled and entangled chains, and when you.
- Apply force, you are straightening them and they get oriented like this.
- So if they are long coiled chains like this entangled, but when you apply force they get stretched like this, therefore they become hard, and then when you release again they go back to the coil form.
- Now presence of polar groups of course is going to result in polymers which will be very stiff and inflexible because of the strong force of attraction which will exist between the chains.
- Now if nonpolar groups are there then it is just weak wanderwald forces and therefore they will be much more.
- We can say elastic in nature and presence of bulky aromatic and cyclic groups of is also going to result in decreasing the elasticity of a polymer.
- If we have an elastic material, can we increase its elasticity?
- That can be done with the help of cross linking as is done in the case of vulcanization.
- What is vulcanization?
- It is treated with sulfur so that the sulfur linkages will be.
- Obtained in between the polymer chains, and since these are covalent bonds, obviously it is going to give much more strength, and when we apply force the chains are not going to slip against each other.
- But if we introduce a small amount of cross linking using sulfur we find that is called balconized rubber and we find that these chains are not going to slip against each other and definitely the tensile strength can be very increased to a good extent.
- Of course we should not introduce too much of vulcanization because instead of becoming elastic it is going to.
- To become stiff because you have increased too much of force of attraction covalent, too many covalent bonds between the chains that is going to result in making it more stiff.
- We have elastic materials.
- But if the material is non-elastic, meaning it's very stiff, can we make it more elastic?
- Yes, that is done with the help of plasticizers.
- What are plasticises?
- They are small organic molecules which are made to enter between the chains.
- Now this can enter between the chains and when it enters between the chains it's understood that the distance between the polymer chains will increase and therefore it will make it less stiff and more flexible.
- The next property is chemical resistance.
- The resistant to resistance to swell, dissolve, and get degraded in the presence of a solvent or a chemical, and it will depend on the chemical nature of the monomeric units which are

being used.
- And their molecular arrangement.
- If there are polar or non-polar groups.
- Because it can easily form hydrogen bonds with water.
- It's all like dissolves like.
- Instead, they will easily swell and sometimes dissolve in non-polar solvents such as petrol, benzene, carbon tetrachloride, etc.
- The second one could be.

Example or application from the lecture

- Suppose this is a polymer chain of natural rubber, what is done in vulcanization?
- These small molecules like for example triphenyl phosphate.
- For example, PVA polyvinyl alcohol will dissolve in water.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 15:00–19:59

is the residual unsaturation.

Key details

- Whenever there is residual unsaturation, it leads to oxidative degradation.
- If there is a double bond in the polymer.
- See addition polymerization when we are doing monomers will always have double bonds.
- That is natural rubber, but when that is the monomer unit in natural rubber.
- But what do we notice here in the polymer itself there is still a double bond.
- When you have such a situation.
- Because of the nature of this double bond.
- In the polymer there is still a double bond, and we know that pi electrons can easily be take part in reaction.
- Therefore, such polymers will undergo oxidative degradation and.
- Because of this residual bond which is there, residual double bond which is there or unsaturation which is there, it gets easily degraded in air and UV light.
- And of course.
- How is dense is the packing?
- Because when there is dense packing in the molecule, then it will not allow the solvent to penetrate between the layers, as in the case of say Teflon.
- We know that Teflon, this is the.
- What happens is they are so close to one another.
- Because of the polarity.
- Each of these bonds is polar in nature.
- CF, CF, CF, all these bonds are polar in nature, and therefore the chains are going to be pulled very close to each other.
- So therefore, it will not allow any solvent.

- To get in between the chains.
- Therefore, we find that Teflon if water cannot enter, so it acts as non-stick.
- Polymer being more chemical resistant.
- That is plastic deformation.
- Some polymers have this property that when we apply heat they become soft and flexible, but on cooling they again return to their original shape.
- This is known as plasticity.
- A thermoplastic polymer would be one which softens on heating and becomes hard on cooling and it can be remoulded into new shapes and that is as the case is of polyethylene, polypropylene, etc.
- But what is a thermosetting polymer it becomes hard on.
- it initially it will be fusible, it becomes fusible, but the moment you cool it first time itself it is going to set into a hard mass and after that it cannot be changed anymore, it cannot be re molded anymore.
- Such polymers are known as thermosetting polymers and it they if at all you heat it then it is going to undergo.
- Total degradation that means it will not melt, rather it is going to degrade into some other compound.
- These are all thermo examples of thermosetting plastics.
- What is the reason?
- Why are they behaving like this?
- It has to do with the structure.
- There could be underwhelm forces, it could be dipole interaction or hydrogen band bonding between the chains.
- So when you are heating then you can easily overcome the force of attraction and therefore you see here when you have a thermoplastin, then when you heat these chains become loose, you have overcome the force of attraction and therefore they start to melt and the moment you reduce the heat or you c cool it, they will come back to the original shape.

Example or application from the lecture

- Like for example isoprene.
- And of course, molecular mass for a given polymer, as the molecular mass is increasing, we find that the swelling character decreases, and of course, if there is decross linking, if there is a high amount of cross-linking in the polymer, the solubility definitely is going to be less.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 20:00–24:59

And in the case of thermosetting, we know that in a thermosetting it's a three-dimensional cross-linked structure which is there.

Key details

- And there are these covalent bonds which are existing between the chains.

- And now when you heat it, we cannot overcome this force.
- Of covalent bond because it's very high.
- Even if you do, then it is going to break into different types of molecules.
- Once these cross link structures are formed, we will not be.
- We are able to get them into.
- get them again remould them again and again.
- Here weak forces of attraction exist whereas here there are covalent bonds and we cannot when we heap them we cannot overcome these strong forces of attraction.
- That is a covalent making the covalent bond.
- What we understand is that the properties of the polymers are dictated by the structure of the polymer.
- So in the next class we are going to talk about some commercial polymers and we will take up butile rubber.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Structure-property relationship transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Structure-property relationship transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Structure-property relationship transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/043_Structure-property_relationship__e79c2e83734020ae.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

C Revision and retrieval practice